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Chemistry
Class 9

Sindh Textbook Board

(English Medium)

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Unit 1
Fundamentals of Chemistry

Unit 2
Structure of Atom

Unit 3
Periodic Table and Periodicity of Properties

Unit 4
Chemical Bonding

Unit 5
Physical States of Matter

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Fundamentals of Chemistry

UNIT 1

1.1 Historical Background of Chemistry

LESSON 1

1. Early Ideas about Matter

The word science comes from the Latin word "Scientia", which means knowledge. Chemistry is the branch of science that deals with the composition, structure, properties and changes of matter. Long ago, Greek thinkers tried to explain what matter is made of.

- **Democritus (460–357 BC, Greek)** presented the idea that matter is made of a smallest, indivisible particle — the atom.
- **Aristotle (428–347 BC, Greek)** presented the theory of the form and structure of matter and gave the idea of four elements: fire, water, earth and air.
- **Plato (428–347 BC, Greek)** suggested the term "element" for the basic constituents of matter.

2. Muslim Scientists

- **Jabir bin Hayyan (721–803 AD)** invented experimental methods for preparing nitric acid and hydrochloric acid, and described methods of dyeing cloth.
- **Al-Razi (862–930 AD)** prepared ethyl alcohol (wine) by the fermentation method.
- **Al-Beruni (973–1048 AD)** determined the densities of different substances.
- **Ibn Sina (980–1037 AD)** presented the sciences of medicine, philosophy and astronomy.

3. Modern Scientists

- **Robert Boyle (1627–1691)** presented the idea that chemistry is a natural science and discovered the gas laws.
- **J. Priestley (1733–1804)** discovered oxygen, sulphur dioxide and hydrogen chloride.
- **Scheele (1742–1786)** discovered chlorine gas.
- **Cavendish (1731–1810)** discovered hydrogen gas.
- **Lavoisier (1743–1794)** discovered that air contains one-fifth (1/5) oxygen.
- **John Dalton (1766–1844)** presented the atomic theory of matter.
- **Avogadro (1776–1856)** presented Avogadro's law about the number of molecules in gases.
- **J.J. Berzelius (1779–1848)** introduced chemical symbols, formulas and chemical equations.
- **Mendeleev (1824–1907)** discovered the periodic arrangement of the elements.

4. Importance of Chemistry in Daily Life

Chemistry serves every field of daily life: cooking and preserving food, medicines, cleaning products, fertilizers and pesticides in agriculture, fuels, clothes, plastics and paints. Every everyday thing is made of atoms and molecules, and its properties depend on its chemical composition; for example food, water and medicines are all chemical substances.

1.2 Branches of Chemistry

LESSON 2

1. What are the Branches of Chemistry?

As knowledge of chemistry increased, it was divided into branches so that each part of the subject could be studied in detail. The main branches of chemistry are:

- **Physical chemistry:** study of the physical properties, changes and energy of matter.
- **Organic chemistry:** study of carbon compounds, especially the compounds of carbon and hydrogen (hydrocarbons).
- **Inorganic chemistry:** study of compounds other than carbon compounds.
- **Analytical chemistry:** study of the composition of matter — identifying its components and measuring their amounts.
- **Biochemistry:** study of chemical processes in living things.

2. Physical Chemistry vs Analytical Chemistry

Physical chemistry studies the physical properties, changes and energy of matter, while analytical chemistry deals with identifying the composition of matter and measuring the amounts of its components.

3. Examples

- Hydrocarbon compounds (carbon and hydrogen) belong to **organic chemistry**.
- Glass, cement and metal-making are connected with industrial applications of chemistry.
- The study of carbohydrates, proteins and fats in living bodies belongs to **biochemistry**.

1.3 Basic Definitions

LESSON 3

1. Matter and Substance

Matter is anything that has mass and occupies space (volume). A **chemical substance** is the pure form of matter with fixed composition and fixed properties, such as an element or a compound.

2. Element, Compound and Mixture

- **Element:** a substance made of only one kind of atom, e.g. hydrogen and oxygen.
- **Compound:** a substance in which elements are joined by chemical bonds in a fixed ratio; its properties are different from its components, e.g. water.
- **Mixture:** a substance in which components are physically mixed in any ratio; it can be separated by physical methods such as filtration, evaporation and distillation, e.g. ice cream and blood.

3. Important Definitions

- **Gram atomic mass:** the mass of one mole of atoms of an element in grams.
- **Gram molecular mass:** the mass of one mole of molecules of a substance in grams.
- **Gram formula mass:** the mass of one mole of formula units of an ionic compound in grams.
- **Atomic mass in grams:** when the atomic mass of an element is expressed in grams it is called gram atomic mass.

4. Chemical, Empirical and Molecular Formulas

The **empirical formula** shows the simplest whole-number ratio of atoms in a compound. The **molecular formula** shows the actual number of atoms in a molecule.

Formula: Molecular formula = $n \times$ empirical formula, where n = molecular mass ÷ empirical formula mass.

Substance	Chemical Formula	Empirical Formula	Molecular Formula
Sulphuric acid	H_2SO_4	H_2SO_4	H_2SO_4
Carbon dioxide	CO_2	CO_2	CO_2
Glucose	$C_6H_{12}O_6$	CH_2O	$C_6H_{12}O_6$
Benzene	C_6H_6	CH	C_6H_6

For glucose, the empirical formula is CH_2O and $n = 6$, so the molecular formula is $C_6H_{12}O_6$. The empirical formula of hydrogen peroxide (H_2O_2) is HO, because the simplest ratio of H to O is 1:1.

5. Identifying Elements, Compounds and Mixtures

- Table salt: compound. Ice cream: mixture. Blood: mixture. Silicon: element. Coca-Cola: mixture. Tin: element. Zinc: element. Sulphur in water: mixture.
- Elements present — washing soda (Na_2CO_3): sodium, carbon, oxygen. Sugar ($C_{12}H_{22}O_{11}$): carbon, hydrogen, oxygen. Sand (SiO_2): silicon, oxygen. Caustic soda ($NaOH$): sodium, oxygen, hydrogen.

1.4 Chemical Species

LESSON 4

1. What is a Chemical Species?

A **chemical species** is a basic particle taking part in a chemical process. The main chemical species are atom, molecule, ion and free radical.

- **Atom:** the smallest particle of an element that takes part in a chemical reaction; it is neutral.
- **Molecule:** a neutral group of two or more atoms joined together.
- **Ion:** a charged atom or group of atoms, formed by gaining or losing electrons; positive (cation) or negative (anion).
- **Free radical:** an atom or group having one unpaired electron, such as $Cl\cdot$ and $CH_3\cdot$. It is very reactive.

2. Types of Molecules (by number of atoms)

- **Monoatomic molecule:** one atom, e.g. He, Ne.
- **Diatomic molecule:** two atoms, e.g. H_2, O_2, Cl_2.
- **Triatomic molecule:** three atoms, e.g. H_2O, CO_2.
- **Polyatomic molecule:** many atoms, e.g. H_2SO_4, $C_6H_{12}O_6$.

3. Homoatomic and Heteroatomic Molecules

A molecule whose atoms are all of the same element is homoatomic, e.g. H2. A molecule containing atoms of different elements is heteroatomic.

4. Atom vs Ion; Molecule vs Molecular Ion; Compound vs Mixture

- An atom is neutral; an ion carries a positive or negative charge.
- A molecule is a neutral group of atoms; a molecular ion is a charged group such as NH4+ and SO4--.
- In a compound the components are joined by chemical bonds in a fixed ratio; in a mixture the components are physically mixed in any ratio.

1.5 Chemical Equations and Balancing Chemical Equations

LESSON 5

1. Chemical Equations

A chemical equation shows a chemical reaction with the help of symbols and formulas. The substances on the left side are reactants and those on the right side are products.

2. Steps of Balancing an Equation

- **First:** write the unbalanced (skeleton) equation.
- **Second:** count the atoms of each element on both sides.
- **Third:** equalise the numbers of atoms with coefficients, but never change the subscripts.
- **Fourth:** re-count at the end to check.

3. Worked Examples

- 4NH3 + 5O2 → 4NO + 6H2O
- 2KNO3 → 2KNO2 + O2
- Ca + 2H2O → Ca(OH)2 + H2
- 2NaHCO3 → Na2CO3 + H2O + CO2
- 2CO + O2 → 2CO2
- CaCO3 + 2HCl → CaCl2 + H2O + CO2
- 4KClO3 → 4KCl + 6O2 (both sides have 4 K, 4 Cl and 12 O atoms)

1.6 Mole and Avogadro's Number

LESSON 6

1. The Mole

A **mole** is the amount of a substance that contains as many particles as there are atoms in 12 g of carbon-12. One mole of any substance contains 6.02 × 10^23 particles.

2. Avogadro's Number

The number 6.02 × 10^23 is called **Avogadro's number (NA)**. One mole of any substance contains Avogadro's number of particles, so the mole and Avogadro's number are directly linked.

3. Molar Mass

The mass of one mole of a substance is its molar mass. For example, the molecular mass of H2SO4 is 98 a.m.u, so its molar mass is 98 g — one mole of H2SO4 weighs 98 gram.

Formula: Number of moles = given mass ÷ molar mass

1.7 Stoichiometry (Chemical Calculations)

LESSON 7

1. Formula Mass and Molecular Mass

Formula mass is the sum of the atomic masses of all atoms in a formula unit. **Molecular mass** is the sum of the atomic masses of all atoms in a molecule.

2. Worked Examples — Formula Mass

- Al2O3 = 2 × 27 + 3 × 16 = 54 + 48 = **102 a.m.u**
- MgCl2 = 24 + 2 × 35.5 = 24 + 71 = **95 a.m.u**
- NaCl = 23 + 35.5 = **58.5 a.m.u**
- KNO3 = 39 + 14 + 3 × 16 = 39 + 14 + 48 = **101 a.m.u**

3. Worked Examples — Molecular Mass

- C2H5OH = 2 × 12 + 6 × 1 + 16 = 24 + 6 + 16 = **46 a.m.u**
- H2O = 2 × 1 + 16 = **18 a.m.u**
- NH3 = 14 + 3 × 1 = **17 a.m.u**
- CO2 = 12 + 2 × 16 = **44 a.m.u**

4. Calculating Moles and Molecules — Full Steps

Example 1: How many moles are needed to get 40 g of H_2SO_4?

Molar mass of H_2SO_4 = 98 g

Moles = given mass ÷ molar mass = 40 ÷ 98 = **0.41 mol**

Example 2: Moles and molecules in 16 g of H_2CO_3:

Molar mass of H_2CO_3 = 62 g

Moles = 16 ÷ 62 = 0.26 mol

Molecules = $0.26 \times 6.02 \times 10^{23}$ = **1.56×10^{23}**

Example 3: Moles and molecules in 20 g of $C_6H_{12}O_6$:

Molar mass of $C_6H_{12}O_6$ = 180 g

Moles = 20 ÷ 180 = 0.11 mol

Molecules = $0.11 \times 6.02 \times 10^{23}$ = **6.7×10^{22}**

Example 4: Moles in 30 g of H_3PO_4:

Molar mass of H_3PO_4 = $3 \times 1 + 31 + 4 \times 16$ = 98 g

Moles = 30 ÷ 98 = **0.31 mol**

Formulas: Moles = given mass ÷ molar mass; Number of particles = moles × 6.02×10^{23}

Structure of Atom

UNIT 2

2.1 Discovery of Subatomic Particles: Electron, Proton and Neutron

LESSON 8

1. Subatomic Particles

The atom is made of three fundamental subatomic particles: electron, proton and neutron. Protons and neutrons are present in the nucleus, while electrons revolve around the nucleus.

Particle	Mass	Charge
Proton	1.67 × 10^-27 kg	+1
Neutron	1.67 × 10^-27 kg	0 (no charge)
Electron	9.1 × 10^-31 kg	−1

2. Atomic Number and Mass Number

- **Atomic number (Z)** = number of protons in the atom. For example, an oxygen atom has 8 protons, so its atomic number is 8.
- **Mass number (A)**, also called **nucleon number**, = total number of protons and neutrons. For example, a chlorine atom with 17 protons and 18 neutrons has mass number 17 + 18 = 35.
- **Number of neutrons** = mass number − atomic number.
- The mass of the atom is shown by the protons and neutrons, which are in the nucleus and make up almost all the mass of the atom.
- The hydrogen atom (protium, 1H) has no neutron.

3. Discovery of the Electron — Cathode Rays

Cathode rays are obtained by passing electricity through a gas at low pressure and high voltage in a discharge tube. Their properties are:

- They travel in straight lines.
- They carry negative charge.
- They bend in electric and magnetic fields.
- They can pass through metal foil.

4. Experiments Proving Electrons, Protons and Neutrons

- **Electron:** the cathode-ray experiment (Crookes and Thomson).
- **Proton:** Goldstein's canal-ray experiment.
- **Neutron:** Chadwick's experiment, in which alpha particles were fired at beryllium and chargeless particles came out.

5. Worked Example

In $_{27}^{59}Co$: protons = 27, electrons = 27, neutrons = 59 − 27 = 32.

2.2 Theories and Experiments Related to Structure of Atom

LESSON 9

1. Rutherford's Gold-Foil Experiment

Rutherford fired alpha particles at a thin gold foil. The alpha particles struck a screen coated with **zinc sulphide (ZnS)** and produced flashes. The observations were:

- Most alpha particles passed straight through the foil.
- Some were deflected from their path.
- A few bounced back.

2. Rutherford's Atomic Model

- The atom is mostly empty space.
- The positive charge and almost all the mass are gathered in a tiny central part called the nucleus.
- Electrons revolve around the nucleus.
- This model is based on the classical idea: electrons can revolve in any orbit.

3. Defects of Rutherford's Model

- It could not explain how the positive charge stays together in the nucleus.
- Orbiting electrons should lose energy and fall into the nucleus, yet the atom is stable.
- It could not explain the line spectrum of the atom.

4. Bohr's Model vs Rutherford's Model

According to Rutherford, electrons can revolve in any orbit, whereas according to Bohr, electrons revolve only in fixed-energy orbits and absorb or release energy when changing orbit. Bohr's model is based on the quantum idea; Rutherford's on the classical idea. In Bohr's model, electrons take in or give out packets (quanta) of energy when they change orbit.

2.3 Modern Theories of Structure of Atom

LESSON 10

1. De Broglie's Hypothesis

In 1923, de Broglie presented the hypothesis that every moving particle also has wave-like properties. The electron too behaves like a wave, so the electron has a dual (wave-particle) nature.

Formula: $\lambda = h/mv$, where h is Planck's constant, m is mass and v is velocity of the particle.

2. Proof of De Broglie's Relation

Planck's equation: $E = h\nu$ and Einstein's equation: $E = mc^2$. Equating them gives $h\nu = mc^2$. Putting $\nu = c/\lambda$ gives $h/\lambda = mc$, so $\lambda = h/mc$. For a moving particle, $\lambda = h/mv$, which is de Broglie's equation.

3. Defects of Bohr's Atomic Model

- It applies only to single-electron particles (like hydrogen).
- It does not explain the fine structure of electron orbits (spectra).
- It goes against Heisenberg's uncertainty principle, because the position and velocity of an electron cannot both be known at the same time.

4. Schrödinger's Atomic Model

According to Schrödinger, electrons are not in fixed orbits but in **orbitals**, where there is a probability of finding them. The structure of the atom is described by the wave equation. This model is based on quantum mechanics.

2.4 Electronic Configuration

LESSON 11

1. Shells and Subshells

- A **shell** is the main energy level of electrons in an atom: K, L, M, N.
- A **subshell** is a smaller division of a shell: s, p, d, f. Example: the L shell contains the 2s and 2p subshells.
- An **s subshell** can hold a maximum of 2 electrons.
- The arrangement (configuration) of electrons belongs to the electron, because electrons are arranged in shells and subshells.
- Electrons fill shells in order because inner shells have lower energy, so electrons first fill the lower-energy shells.

2. Working Out Configurations

- If the K and L shells are completely full, the atom has 2 + 8 = 10 electrons in total.
- Potassium (19 electrons): 2, 8, 8, 1.
- Sodium (Z = 11): 2, 8, 1 — so its L shell has 8 electrons.

3. Worked Example — Atom with 5 Electrons in the M Shell

(a) Atomic number = 2 + 8 + 5 = 15

(b) Electronic configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$, or 2, 8, 5

(c) Element: phosphorus (P)

4. Counting Protons, Neutrons and Electrons

Atom	Protons	Electrons	Neutrons
56Fe26	26	26	30
1708	8	8	9
37Cl17	17	17	20
235U92	92	92	143
14C6	6	6	8

Potassium has nucleon number 39 and proton number 19, so its neutrons = 39 − 19 = 20.

2.5 Isotopes and Their Common Uses

LESSON 12

1. What are Isotopes?

Isotopes are atoms of the same element with the same proton number but different nucleon numbers, i.e. they differ in the number of neutrons.

2. Isotopes of Chlorine

- 35Cl17: 17 protons, 18 neutrons.
- 37Cl17: 17 protons, 20 neutrons.

3. Isotopes of Hydrogen

- Protium: 1 proton, 0 neutrons.
- Deuterium: 1 proton, 1 neutron.
- Tritium: 1 proton, 2 neutrons.

4. Isotopes of Carbon and Oxygen

All isotopes of carbon have 6 protons, with 6, 7 or 8 neutrons. The abundance of the carbon-12 isotope is about 98.9%. 160 and 170 are alike because both have 8 protons and 8 electrons and the same chemical properties; they differ because 160 has 8 neutrons and 170 has 9 neutrons, so their masses and physical properties differ.

5. Why Same Chemical but Different Physical Properties?

Chemical properties depend on the electron arrangement, which is the same in isotopes, while physical properties depend on the mass number, which differs.

6. Uses of Isotopes in Daily Life

- Carbon-14 is used to date old objects.
- Cobalt-60 is used in cancer treatment; its gamma rays also sterilise medical instruments because they carry very high energy and kill germs.
- Iodine-131 is used to examine thyroid disease.
- Uranium-235 is used as fuel in nuclear power stations.

Periodic Table and Periodicity of Properties

UNIT 3

3.1 Periodic Table

LESSON 13

1. What is the Periodic Table?

The periodic table is an ordered list of the elements in which the elements are arranged in increasing atomic number. It has 7 periods (horizontal rows) and 18 groups (vertical columns).

2. Early Attempts and Mendeleev's Periodic Law (1869)

Mendeleev presented the periodic law that the properties of elements are a periodic function of their **atomic mass**. He arranged the elements in increasing order of atomic mass and grouped elements of similar properties together. He left gaps for undiscovered elements and predicted their properties, which later proved correct. Lothar Meyer also presented a similar table of 56 elements in 1869.

3. Modern Periodic Law — Moseley (1913)

Moseley discovered that the atomic number is the basic property of an element. According to the modern periodic law, the chemical properties of elements are a periodic function of their **atomic number**. The atomic number fixes the element's place in the table, its electronic configuration and its properties — so the periodic properties of an element depend on the number of protons in its atom.

4. Periods and Groups

- A **period** is a horizontal row of the periodic table in which elements are in increasing atomic number.
- A **group** is a vertical column whose elements have equal electrons in the outer shell and similar properties.
- The 4th and 5th periods are called long periods because they contain 18 elements each.

5. Blocks of the Periodic Table

The table is divided into s, p, d and f blocks on the basis of electronic configuration, i.e. which subshell the last electron enters:

Block	Groups	Last electron enters
s block	IA and IIA	s subshell
p block	IIIA to VIIIA	p subshell
d block	Transition elements	d subshell
f block	Lanthanides and actinides	f subshell

6. Families of Elements

- Group IA: alkali metals (Li, Na, K).
- Group IIA: alkaline-earth metals (Mg, Ca, Ba).
- Groups IIIA to VIA: representative elements.
- Group VIIA: halogens (F, Cl, Br, I) — fluorine and chlorine are gases, bromine is a liquid and iodine is a solid.
- Group VIIIA: noble (inert) gases.
- Groups IB to VIIIB: transition metals — the transition and inner-transition elements are metals, such as iron, copper and zinc.
- Lanthanides: from lanthanum (La) to lutetium (Lu). Actinides: from actinium (Ac) to lawrencium (Lr).

7. Reading the Table

- The atomic number is written at the top of each element's cell; for example carbon has atomic number 6 and sodium 11.
- Each cell carries the element's symbol and name; for example H is hydrogen, O is oxygen, Fe is iron and Au is gold.
- Most metals are solids with high melting points, bromine is a liquid, and non-metals such as hydrogen and oxygen are gases.
- Elements of the same group (family) have equal electrons in the outer shell, so their chemical properties are similar and their physical properties change in order; e.g. the alkali metals Li, Na, K are all soft and reactive.

8. Electronic Configurations (examples)

- Na ($Z = 11$): $1s^2\ 2s^2\ 2p^6\ 3s^1$ or 2, 8, 1
- Ca ($Z = 20$): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$ or 2, 8, 8, 2
- F ($Z = 9$): $1s^2\ 2s^2\ 2p^5$ or 2, 7
- Si ($Z = 14$): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$ or 2, 8, 4

3.2 Periodicity of Properties

LESSON 14

1. Periodic Properties

Properties that change in a regular way across a period and down a group are called periodic properties. The main ones are atomic radius, ionisation energy, electron affinity and electronegativity.

2. Atomic Radius

- Across a period (left to right), the atomic radius **decreases**, because the nuclear pull increases while the shells stay the same.
- Down a group, the atomic radius **increases**, because new shells keep being added.

3. Shielding Effect

The **shielding effect** is that the inner electrons hide (block) the nuclear pull from the outer electrons, so the outer electrons feel less pull. The greater the shielding, the weaker the nuclear effect and the larger the atomic radius. Across a period the shielding effect decreases in effect, because the nuclear charge increases while the inner electrons stay the same.

4. Ionisation Energy, Electron Affinity and Electronegativity

- **Ionisation energy:** the energy needed to remove an electron from an atom. Helium has the highest ionisation energy, because its atom is the smallest and its electrons are held very tightly by the nucleus.
- **Electron affinity:** the energy change when an electron is added to an atom. It decreases down a group because the atomic size increases, so the nuclear pull on the outer electron decreases.
- **Electronegativity:** the attraction of an atom for a shared pair of electrons in a bond.

5. Trends in a Group and a Period

Property	Down a group	Across a period (left → right)
Atomic radius	Increases	Decreases
Ionisation energy	Decreases	Increases
Electron affinity	Decreases	Increases
Electronegativity	Decreases	Increases

Electronegativity examples: down a group it decreases (F = 4.0, Cl = 3.0, Br = 2.8); across a period it increases (Li = 1.0, C = 2.5, F = 4.0). Ionisation energy, electron affinity and electronegativity all increase across a period because the nuclear pull increases and the atomic size decreases.

Note: magnesium (Mg), an alkaline-earth metal, forms a protective oxide coating on its surface, which protects the metal underneath.

Chemical Bonding

UNIT 4

4.1 Why Do Atoms Form Chemical Bonds?

LESSON 15

1. Why Do Atoms Combine?

Atoms form chemical bonds to gain stability. Elements gain stability by completing their outer shell — by giving, taking or sharing electrons. Metals give electrons; non-metals take or share electrons, so that the octet or duplet is completed.

2. Octet Rule and Duplet Rule

- **Octet rule:** an atom tends to complete 8 electrons in its outer (valence) shell to become stable.
- **Duplet rule:** small atoms such as hydrogen and helium tend to complete 2 electrons in their shell.

3. Why are Noble Gases Stable?

Noble gases have 8 electrons in their valence shell (2 in helium), so their shells are complete. Therefore they need neither to gain nor to lose electrons, they do not react with other elements, and they are stable. For the same reason, a helium atom does not gain electrons: its shell already has 2 electrons and its duplet is complete.

4. Valence Shell and Valence Electrons

The outer shell of an atom is its valence shell, and it can hold at most 8 electrons. The electrons in this shell are called valence electrons. They are important because they take part in bonding and chemical reactions. For example, magnesium has 2 electrons in its valence shell (2, 8, 2).

4.2 Formation of Chemical Bonds

LESSON 16

1. How Chemical Bonds Form

Atoms form bonds by giving, taking or sharing their valence electrons so that each atom completes its octet or duplet. Metals give electrons and form cations; non-metals gain electrons and form anions, or share electrons.

2. Ions and Atoms

An **ion** is a charged atom or group of atoms. An atom is neutral, whereas an ion forms by gaining or losing electrons and is positive (cation) or negative (anion).

3. Formation of Cations (metals give electrons)

- K (2, 8, 8, 1) → K+ (2, 8, 8) + e−. A Group IIIA atom gives up 3 electrons; magnesium gives 2 electrons.
- Al (2, 8, 3) → Al3+ (2, 8) + 3e−.
- Magnesium's arrangement is 2, 8, 2. It readily gives two electrons because by giving them it becomes Mg2+ and gains a stable octet. Magnesium is in Group IIA and its ion carries a +2 charge, symbol Mg2+.
- An atom with 2 electrons in its valence shell: magnesium (Mg), arrangement 2, 8, 2.

4. Formation of Anions (non-metals gain electrons)

- Oxygen (2, 6) gains two electrons to form O2− and completes its octet: O + 2e− → O2− (arrangement 2, 8).
- S + 2e− → S2− (arrangement 2, 8, 8).
- N + 3e− → N3−; Br + e− → Br−; P + 3e− → P3−.
- A non-metal atom gets a negative charge by gaining electrons, e.g. Cl + e− → Cl−.
- The fluoride ion (F−) has 10 electrons, the same as neon (Ne), so its arrangement is neon-like. By sharing one electron in bonding, a fluorine atom also gets the configuration of neon.

5. Dot-and-Cross Diagrams

- **H2 (covalent):** H· + ·H → H:H; both H atoms share electrons.
- **KF (ionic):** K (2, 8, 8, 1) gives its one electron to F (2, 7); K+ and F− form and join by attraction.

Z	Element	Protons	Electrons	Arrangement	Valence electrons
11	Na	11	11	2, 8, 1	1
12	Mg	12	12	2, 8, 2	2
13	Al	13	13	2, 8, 3	3
14	Si	14	14	2, 8, 4	4
15	P	15	15	2, 8, 5	5
16	S	16	16	2, 8, 6	6

4.3 Types of Chemical Bonds

LESSON 17

1. Ionic Bond

An ionic bond forms by the **transfer of electrons**: one atom gives electrons, the other takes them; ions form and join by the attraction of opposite charges. Ionic compounds are compounds formed by the attraction between positive and negative ions, such as NaCl and MgO.

Formation of NaCl: Na (2, 8, 1) gives its one electron to become Na+; Cl (2, 8, 7) takes an electron to become Cl−. The attraction of opposite charges forms NaCl.

2. Covalent Bond

A covalent bond forms by **sharing electron pairs**, so that both atoms complete their octets.

- **Single covalent bond:** one shared pair, e.g. H2, Cl2.
- **Double covalent bond:** two shared pairs.
- **Triple covalent bond:** three shared pairs, e.g. N2.
- **Non-polar covalent bond:** the electron pair is shared equally, e.g. H2 and Cl2 (atoms of equal electronegativity show no + or − marks).
- **Polar covalent bond:** the pair is pulled more towards the more electronegative atom; polarity occurs in molecules such as HCl, H2O and NH3.

3. Coordinate Covalent Bond

A coordinate covalent bond is a bond in which the shared electron pair is given by only one atom and accepted by the other. Examples: in NH4+, nitrogen gives its lone pair to H+; in H3O+, oxygen gives its lone pair to H+. A coordinate covalent bond is always a polar bond, because the pair comes from one atom only, so partial charges separate in the bond.

4. Metallic Bond

A metallic bond is the bond between metal atoms in which the valence electrons are shared by all the atoms and form a **sea of electrons**. Positive metal ions are held in this sea.

5. Lone Pair and Bond Pair

A **lone pair** does not take part in bonding and stays on one atom. A **bond pair** (shared pair) is shared between two atoms and forms a bond — it is the electron pair an atom contributes in bonding.

6. Bond Nature from Electronegativity Difference

Electronegativity difference	Type of bond
0 to 0.4	Non-polar covalent
0.4 to 1.7	Polar covalent
More than 1.7	Ionic bond

When the electronegativity of two atoms differs, the shared pair is pulled towards the more electronegative atom, so the bond gains partial **ionic character**. The greater the difference, the greater the ionic character. When opposite partial charges appear on the two sides of a bond, that bond becomes dipolar (polar).

7. Why Metals Form Ionic but Not Covalent Bonds

Metals like to give electrons, so they form ions and make ionic bonds. A covalent bond requires sharing electrons, which is not the nature of metals.

8. Dot-Cross Structures of Molecules

- **H2O:** O in the centre, single bonds to two H atoms, two lone pairs on O.
- **N2:** N≡N triple bond, one lone pair on each N.
- **CH4:** C in the centre, single bonds to 4 H atoms.
- **C2H2:** H−C≡C−H.
- **Cl2:** Cl−Cl single bond. **H2:** H−H single bond.
- **CCl4:** carbon in the centre forming single bonds to 4 chlorines; each chlorine has 3 lone pairs and carbon's

octet is complete.
- A hydrogen atom cannot form more than one covalent bond, because it has only 1 electron and needs only one more electron to complete its duplet.

9. Polar vs Non-Polar Compounds

- **Polar compounds:** partially charged, soluble in water, higher melting and boiling points.
- **Non-polar compounds:** uncharged, insoluble in water, lower melting and boiling points.

4.4 Intermolecular Forces

LESSON 18

1. What are Intermolecular Forces?

The **intermolecular force** is the weak attraction that holds neighbouring molecules together. It is weaker than the intramolecular force because intramolecular forces are strong chemical bonds between atoms, whereas intermolecular forces are only weak attractions between molecules.

2. Hydrogen Bond

The mutual interaction between a hydrogen atom bonded to a highly electronegative atom and another electronegative atom is called a **hydrogen bond**. Hydrogen bonding holds molecules close together, so:
- Melting and boiling points rise.
- Solubility in water increases.
- Special properties, such as those of water, appear.

3. Dipole-Dipole Force

The dipole force is the attraction between polar molecules: the positive end of one molecule attracts the negative end of another molecule.

4.5 Nature and Properties of Bonds

LESSON 19

1. Properties of Ionic Compounds

- They are crystalline solids at room temperature.
- They have high melting and boiling points, because the attraction between ions is very strong.
- They conduct electricity in the molten or dissolved state, but not as solids.
- They are hard and brittle.
- In the NaCl crystal, each Na^+ ion is surrounded by 6 Cl^- ions.

2. Why Covalent Compounds Have Low Melting Points

The covalent bonds inside a molecule are strong, but the forces between molecules are weak. On melting, the weak forces between molecules break, not the inner bonds, so the melting point is low — e.g. ice. Compared with an ionic bond, a covalent bond's effect is weaker in this sense: the ionic bond is stronger because of the strong attraction between ions.

3. Properties of Metallic Bonds

In a metallic bond, metal atoms are joined in a sea of electrons. That is why metals are:
- Good conductors of heat and electricity — metals have free/shared electrons which move easily and carry heat and electricity from one place to another.
- Shiny (lustrous).
- Malleable (can be beaten into sheets) and ductile (can be drawn into wires).

4. Adhesives and Epoxy Adhesives

Adhesives are used to join broken things, to stick wood, paper, cloth and plastic, and to join parts in electronics — where electrical insulation/barrier is needed. Epoxy adhesives are very strong and are used in electronics, aircraft and buildings. Adhesives are important in society for daily repairs and for joining parts in industry.

Physical States of Matter

UNIT 5

5.1 Gaseous State

LESSON 20

1. The Gaseous State

In a gas, the particles are very far apart and move freely in all directions, so a gas has no fixed shape or volume and its density is low — gas is the lightest form of matter among solid, liquid and gas.

2. Kinetic Molecular Theory and the States of Matter

- In a **solid**, the particles are close and only vibrate in place.
- In a **liquid**, the particles can flow over one another.
- In a **gas**, the particles are far apart and move freely.
- As temperature rises, the motion of the particles increases.

3. Diffusion and Effusion

- **Diffusion:** the spontaneous spreading of gas or liquid molecules due to random motion, such as perfume spreading in a room. Lighter gases spread faster.
- **Effusion:** gas escaping through a tiny hole, such as gas leaking from a balloon.
- Both depend on molar mass (Graham's law), temperature and pressure. Hydrogen diffuses fastest because it is the lightest gas (molecular mass = 2).

4. Effect of Heat on a Gas

On heating, a gas expands and its volume increases, so its density decreases.

5.2 Gas Laws

LESSON 21

1. Pressure and Temperature Units

- One atmosphere pressure = 101325 pascal (Pa).
- Temperature conversion: K = °C + 273; °C = K − 273.
- Absolute zero is −273°C (0 K), the temperature at which the volume of a gas would become zero and molecular motion almost stops.

2. Conversions — Worked Examples

- 100°C = 100 + 273 = 373 K
- 150°C = 150 + 273 = 423 K
- 750 K = 750 − 273 = 477°C
- 170 K = 170 − 273 = −103°C

3. Boyle's Law

At constant temperature, the volume of a gas is inversely proportional to its pressure.

Formula: PV = K, i.e. P1V1 = P2V2. In Boyle's law, pressure (P) and volume (V) are the variables, while temperature stays constant.

Example: pressing air in a syringe decreases its volume and increases its pressure.

Example 1: A gas at STP occupies 40.0 dm³; at constant temperature the pressure becomes 3 atm. New volume?

P1V1 = P2V2 → V2 = 1 × 40.0 ÷ 3 = **13.33 dm³**

Example 2: A gas of 800 cm³ at 750 mm Hg is compressed to 250 cm³. New pressure?

P1V1 = P2V2 → P2 = 800 × 750 ÷ 250 = **2400 mm Hg**

Example 3: A gas sample at 8 atm has volume 15 litres; pressure is brought to 6 atm. New volume?

P1V1 = P2V2 → V2 = 8 × 15 ÷ 6 = **20 litres**

4. Charles's Law

At constant pressure, the volume of a gas is directly proportional to its absolute temperature. When temperature increases, a gas expands (at constant pressure).

Formula: V/T = K, i.e. V1/T1 = V2/T2. On a graph, the relation between volume and temperature is a straight line.

Example: warming a balloon makes it expand.

Example 1: A gas expands from 90.5 cm³ to 120 cm³ at constant pressure; initial temperature is 33°C. Final temperature?

T1 = 33 + 273 = 306 K

$T2 = V2 \times T1 \div V1 = 120 \times 306 \div 90.5 = 405.6\ K =$ **132.6°C**

Example 2: A gas of 78 ml at constant pressure is heated from 35°C to 80°C. Final volume?

$T1 = 35 + 273 = 308\ K$; $T2 = 80 + 273 = 353\ K$

$V2 = V1 \times T2 \div T1 = 78 \times 353 \div 308 =$ **89.4 ml**

5.3 Liquid State

LESSON 22

1. The Liquid State

In a liquid, the particles are close but can flow over one another, so a liquid has a fixed volume but no fixed shape.

2. Evaporation

Evaporation is the process in which liquid molecules escape from the surface as vapour at any temperature. The molecules that leave are those which have **more energy**, so they escape the intermolecular attraction and leave the surface.

- **Factors affecting evaporation:** temperature, surface area, air movement, intermolecular forces and the nature of the liquid. External pressure does not affect the rate of evaporation much, because evaporation happens at the surface.
- Evaporation increases with temperature: as temperature rises, more molecules gain enough energy to escape from the surface as vapour.
- **Evaporation causes cooling** because the higher-energy molecules leave and the average energy of the remaining molecules falls.

3. Vapour Pressure

Vapour pressure is the pressure of a liquid's vapour above the liquid. It occurs in a closed system: the vapour cannot escape, so equilibrium is set up between evaporation and condensation and the pressure becomes constant — this is called dynamic equilibrium, in which the rate of evaporation becomes equal to the rate of condensation. The vapour pressure of water increases with temperature, because as temperature rises more molecules become vapour.

4. Boiling Point

- **Boiling point:** the temperature at which the vapour pressure of a liquid equals atmospheric pressure and the liquid starts to boil.
- The boiling point depends on pressure: when atmospheric pressure rises, the boiling point rises; when it falls, the boiling point falls — for example, on mountains water boils below 100°C.
- **Freezing point:** the temperature at which a liquid becomes a solid. **Melting point:** the temperature at which a solid becomes a liquid.

5. Diffusion in Liquids

Diffusion in liquids is the spreading of the molecules of one liquid through another, such as ink spreading in water. It is affected by temperature, particle size and the density of the liquid. The rate of diffusion in liquids is lower than in gases because in a liquid the molecules are close and held by strong attraction, so they move slowly — ink spreads slowly in water, while perfume spreads quickly in air.

5.4 Solid State

LESSON 23

1. The Solid State

In a solid, the particles are packed close together and only vibrate in place — that is why the particles of a solid are the slowest-moving, and a solid has a definite shape and volume.

2. Effect of Melting Points on Solids

- Solids with **higher melting points** have stronger bonds and melt slowly on heating, such as diamond.
- Solids with **lower melting points** melt quickly, such as ice and wax.

So the melting point of a solid shows the strength of the forces holding its particles: strong forces need more heat to break, and weak forces break with little heat.

5.5 Types of Solids

LESSON 24

1. Types of Solids

Crystalline solids	Amorphous solids
Particles arranged in a regular, orderly pattern	Particles arranged in disorder
Have a definite (sharp) melting point	Have no definite melting point; soften gradually
Examples: diamond, salt	Examples: glass, plastic, rubber

Diamond is a crystalline solid, so it has a definite melting point. Glass, plastic and rubber are amorphous solids and do not melt at one sharp temperature.

5.6 Allotropy

LESSON 25

1. What is Allotropy?

Allotropy is the property in which the same element occurs in different forms. These different forms of an element are called allotropes. They have the same chemical element but different arrangements of atoms, so their physical properties differ.

2. Allotropes of Carbon

- **Diamond:** a hard crystalline form of carbon with a definite melting point.
- **Graphite:** another crystalline form of carbon, different in structure from diamond.
- **Fullerenes:** a third form of carbon with cage-like molecules.

5.7 Plasma State

LESSON 26

1. What is Plasma?

Plasma is the fourth state of matter. In plasma, the gas atoms split into ions and free electrons because of very high temperature or energy. It is a mixture of positive ions and free electrons.

2. Examples of Plasma in Daily Life

- Lightning in the sky.
- Neon lights and sign tubes.
- TV screens (plasma screens).
- Plasma lamps.

5.8 Bose-Einstein Condensate

LESSON 27

1. What is Bose-Einstein Condensate?

Bose-Einstein Condensate (BEC) is a state of matter that forms near absolute zero (0 K, −273°C). It was presented as an idea by S.N. Bose and Albert Einstein, and was first prepared by Eric Cornell, with Carl Wieman making its arrangement.

2. Why is it Super-Cold and Super-Unexcited?

- It forms near absolute zero, 0 K = −273°C, so it is **super-cold**.
- At such cold, the motion of the atoms almost stops, so it is **super-unexcited** matter — the atoms have almost no energy of motion.
- It is the opposite extreme of plasma: plasma forms at very high energy, while BEC forms at almost zero energy.

Solutions

UNIT 6

6.1 Solutions, Aqueous Solutions, Solute and Solvent

LESSON 28

1. What is a Solution?

A **solution** is a homogeneous mixture formed when a solute dissolves in a solvent. A solution is called a mixture because two or more substances are uniformly mixed in it, without a chemical reaction.

2. Components of a Solution

- **Solute:** the component present in the smaller amount, which dissolves, e.g. salt in salt solution.
- **Solvent:** the component present in the larger amount, which dissolves the solute, e.g. water in salt solution.
- An **aqueous solution** is a solution in which the solvent is water.
- In air, nitrogen is the solvent because it is present in the largest amount (about 78%); oxygen and carbon dioxide are solutes.

3. Properties of a Solution

- It is a homogeneous mixture.
- Its particles are very small (smaller than 1 nm).
- Its particles do not settle down.
- Light passes through it without scattering.

4. How Salt Dissolves in Water

When salt is added to water, the polar water molecules attract the Na^+ and Cl^- ions. The oxygen ends of water surround Na^+ and the hydrogen ends surround Cl^-, so the salt crystals break apart and dissolve. This process is called **hydration**. The oxygen end of water is partially negative and the hydrogen end is partially positive.

5. Importance of Solutions in Life

Most processes in our body and daily life happen in solution, such as food digestion, medicines working and cleaning. For a chemist, solutions are useful because in solutions chemical reactions happen quickly, solutions can be measured, standard solutions can be made, and experiments such as titration can be performed.

6.2 Saturated, Unsaturated and Supersaturated Solutions

LESSON 29

1. Saturated Solution

A **saturated solution** holds the maximum amount of solute at that temperature; extra solute added to it stays undissolved.

2. Unsaturated Solution

An **unsaturated solution** can still dissolve more solute. It is prepared by dissolving less solute than the solubility limit. Heating a saturated KCl solution makes it unsaturated, because on heating more solute can dissolve.

3. Supersaturated Solution

When a saturated solution is cooled, it can become **supersaturated**: cooling lowers the solubility and the extra solute stays dissolved in the solution. A supersaturated solution is prepared by heating a saturated solution to dissolve more solute, then cooling it slowly without disturbance.

4. Beaker Experiment

Beaker A (20 mL water + 10 g sodium thiosulphate, all dissolved) has the unsaturated solution, while beaker B, with more solute (20 g) and undissolved solid remaining, has the saturated solution.

6.3 Types of Solutions and Dilution of Solutions

LESSON 30

1. Types of Solutions (by state of solute and solvent)

Type	Example
Solid in liquid	Salt in water (salt solution)
Gas in gas	Air (oxygen and other gases in nitrogen)
Solid in solid	Brass (zinc in copper); alloys
Liquid in solid	Amalgam (a metal in mercury)
Liquid in gas	Fog/mist (water droplets in air)
Liquid in liquid	Alcohol in water (alcohol solution)

- An **alloy** is a homogeneous mixture of two or more metals — a solid-in-solid solution, e.g. brass: solute = zinc (smaller amount, 20%), solvent = copper (larger amount, 80%).
- A heterogeneous mixture is one that is not uniform, e.g. water and sand, because sand does not dissolve in water.

2. Dilute and Concentrated Solutions

A **dilute solution** has little solute, whereas a **concentrated solution** has much solute. A concentrated solution can be made from a dilute solution by evaporating solvent or by adding more solute.

3. Dilution

Dilution means making a solution more dilute, i.e. adding solvent to it.

Formula: $M1V1 = M2V2$

Example: How much 4M stock solution is needed to make 200 ml of 0.5M solution?

$M1V1 = M2V2 \rightarrow 4 \times V1 = 0.5 \times 200 \rightarrow V1 = 100 \div 4 =$ **25 ml**

Example: Preparing 500 cm³ of 0.1M HNO_3 from 10M HNO_3:

$10 \times V1 = 0.1 \times 500 \rightarrow V1 =$ **5 cm³**; take 5 cm³ of the 10M acid and dilute with water to 500 cm³.

6.4 Units of Concentration of Solutions

LESSON 31

1. Concentration of a Solution

Concentration is the amount of solute in a given amount of solution or solvent. It can be expressed as percent or as molarity. A solution with lower molarity is more dilute: 2M is more dilute than 3M, and 0.25M has the least solute among 0.5M, 0.75M, 1.5M and 0.25M.

2. Percent Concentrations

- **(m/m)%:** grams of solute in 100 g of solution. A 10% (m/m) sugar solution means 10 g sugar in 100 g solution, i.e. 10 g sugar + 90 g water.
- **(m/v)%:** grams of solute in 100 cm³ of solution. Example: 5% (m/v) means 5 g solute in 100 cm³ of solution.
- **(v/v)%:** volume of solute in 100 cm³ of solution, e.g. 20 cm³ alcohol + 80 cm³ water = 20% (v/v).

3. Molarity

Molarity is the number of moles of solute in 1 dm³ of solution. A **molar solution** is one in which 1 mole of solute is dissolved in 1 dm³ of solution, called 1M. Example: dissolving 40 g NaOH in 1 dm³ of solution makes a 1M NaOH solution.

Formulas: Molarity = moles of solute ÷ volume of solution in dm³; Mass = molarity × volume (dm³) × molar mass

4. Worked Examples

Example 1: Molarity of 40 g NaCl in 500 cm³ solution: moles = 40 ÷ 58.5 = 0.684; volume = 0.5 dm³; M = 0.684 ÷ 0.5 = **1.37M**

Example 2: 1.25 g HCl made to 30 cm³: molar mass HCl = 36.5 g; moles = 1.25 ÷ 36.5 = 0.03425; volume = 0.030 dm³; molarity = 0.03425 ÷ 0.030 = **1.14M**

Example 3: 2.5 g KCl in 100 cm³: molar mass KCl = 74.5 g; moles = 2.5 ÷ 74.5 = 0.03356; concentration = 0.03356 ÷ 0.1 = **0.34 mol/dm³**

Example 4: NaOH in 1 dm³ of 0.5M solution: moles = 0.5 × 1 = 0.5; mass = 0.5 × 40 = **20 g**

Example 5: NaOH needed for 400 cm³ of 0.3M: moles = 0.3 × 0.4 = 0.12; mass = 0.12 × 40 = **4.8 g**

Example 6: 100 cm³ mineral water contains 28 g calcium: concentration = 28 ÷ 0.1 = **280 g/dm³**

Example 7: % (m/m) of 10 g sugar in 140 g water: total = 150 g; % = 10 ÷ 150 × 100 = **6.67%**

Example 8: A 1.2M NaOH solution: mass in 1 dm^3 = 1.2 × 40 = 48 g/dm^3; in 500 cm^3 = **24 g**

2 moles of NaCl = 2 × 58.5 = 117 g (molar mass of NaCl = 58.5 g).

6.5 Solubility

LESSON 32

1. What is Solubility?

Solubility is the maximum amount of solute that dissolves in 100 g of solvent at a given temperature. When no more solute can dissolve in the solution, the solution is saturated and that is the limit of concentration.

2. Effect of Temperature on Solubility

The solubility of most solid solutes, such as salt, increases with temperature, because a hot solvent can dissolve more solute.

3. "Like Dissolves Like"

A solute dissolves because the solute-solvent attraction is stronger than the solute-solute and solvent-solvent forces. The general rule is **"like dissolves like"**:

- A polar solute dissolves in a polar solvent, and a non-polar solute dissolves in a non-polar solvent, because molecules with the same kind of intermolecular forces attract one another. Example: salt in water, grease in benzene.
- Polar solutes dissolve only in polar solvents because the molecules of a polar solute attract the molecules of a polar solvent.
- A polar solute cannot dissolve in a non-polar solvent, because the non-polar solvent cannot separate the strongly attracting polar solute particles.
- Petrol is non-polar and water is polar, so petrol does not dissolve in water. Benzene is non-polar, so it also does not dissolve in water.
- A non-polar solute does not dissolve in water because the polar water molecules attract one another strongly by hydrogen bonding, and a non-polar solute cannot break in between them.
- If the solute-solvent attraction is stronger than the solute-solute attraction, a solution can form, because the solvent separates the solute particles and surrounds them.

6.6 Comparison of Solution, Suspension and Colloids

LESSON 33

1. Solution, Colloid and Suspension — Comparison

Feature	Solution	Colloid	Suspension
Particle size	Smaller than 1 nm	1 to 1000 nm	Larger than 1000 nm
Homogeneity	Homogeneous	Less homogeneous; scatters light	Heterogeneous
Settling	Particles do not settle	Particles do not settle	Particles settle down
Light	Passes through	Scatters light (Tyndall effect)	Scatters/blocks light

2. Colloids and the Tyndall Effect

- The particles of a colloid scatter light, so the beam of light is visible in the colloid. This is called the **Tyndall effect**, like a car's light being visible in fog.
- Colloid particles are very small and always in motion, so they do not settle at the bottom.
- A colloidal solution is shaken well before use so that its particles spread evenly again.
- Examples: milk (used as food), fog, starch solution. Glucose solution is a true solution, while starch solution is a colloidal solution.
- A false (non-true) solution is one whose particles are larger than true-solution particles and which scatter light, such as a colloid or suspension.

Electrochemistry

UNIT 7

7.1 Oxidation and Reduction

LESSON 34

1. Oxidation and Reduction

- **Oxidation:** losing electrons or gaining oxygen, e.g. $Zn \rightarrow Zn^{2+} + 2e^-$.
- **Reduction:** gaining electrons or losing oxygen, e.g. $Cl_2 + 2e^- \rightarrow 2Cl^-$.
- Oxidation and reduction always happen together in a redox reaction.

2. Oxidising and Reducing Agents

- **Oxidising agent:** the substance that takes electrons and oxidises another substance, e.g. Cl_2, H_2SO_4, $KMnO_4$.
- **Reducing agent:** the substance that gives electrons and reduces another substance, e.g. Zn, Al, Na, H_2S.
- The oxidising agent itself is reduced, and the reducing agent itself is oxidised.

3. Identifying Agents in Equations

Equation	Oxidising agent (reduced)	Reducing agent (oxidised)
$Zn + Cl_2 \rightarrow ZnCl_2$	Cl_2	Zn
$Br_2 + H_2S \rightarrow 2HBr + S$	Br_2	H_2S
$2Ca + O_2 \rightarrow 2CaO$	O_2	Ca
$2Li + S \rightarrow Li_2S$	S	Li
$H_2 + Cl_2 \rightarrow 2HCl$	Cl_2	H_2

From the list Al, Na, H_2S, H_2SO_4, $KMnO_4$, Zn: oxidising agents are H_2SO_4 and $KMnO_4$; reducing agents are Al, Na, H_2S and Zn.

7.2 Electrochemical Cells

LESSON 35

1. Electrolytes

- An **electrolyte** is a substance that conducts electricity in the molten or aqueous state through free ions.
- A **strong electrolyte** dissociates fully into ions in water, e.g. HCl, NaOH, KCl.
- A **weak electrolyte** ionises only slightly in water, e.g. CH_3COOH (acetic acid), H_2S, NH_4OH.
- A **non-electrolyte** does not produce ions and does not conduct electricity, e.g. glucose, urea, benzene. Urea is not an electrolyte.
- Graphite is not an electrolyte because it conducts electricity through electrons, not ions.
- Only ionic compounds conduct electricity in the molten or aqueous state, because in these states their free ions can move and carry charge. In the solid state the ions are locked in place, so electricity cannot flow.

2. Electrochemical Cells

- A **voltaic (galvanic) cell** converts chemical energy into electrical energy, such as the Daniel cell. An **electrolytic cell** uses electrical energy to drive a chemical reaction, such as the electrolysis of water.
- Both have two electrodes (anode and cathode) and an electrolyte.
- In a Daniel cell, zinc (Zn) is the anode and copper (Cu) is the cathode.
- **Reduction occurs at the cathode and oxidation at the anode.** In an electrolytic cell, oxidation occurs at the anode (+) and reduction at the cathode (−); electrons flow from the anode to the cathode through the external circuit. K^+ ions move to the cathode and Cl^- ions to the anode in the electrolysis of KCl solution.

3. Dry Cell

In a dry cell, the zinc cover is the anode, the carbon rod in the middle is the cathode, and NH_4Cl paste between them is the electrolyte. A chemical reaction produces electricity. It is used in torches, radios and toys.

4. Lead Storage Battery

A lead storage battery is a rechargeable battery with lead plates and sulphuric acid. On discharge, lead sulphate forms; on charging, lead and lead dioxide form again. It is used in vehicles.

5. Faraday's Laws of Electrolysis

- **First law:** the mass of substance deposited at an electrode is directly proportional to the quantity of electricity passed.
- **Second law:** when the same quantity of electricity is passed, the masses of different substances deposited are proportional to their equivalent weights.
- One Faraday = 96500 coulombs of charge.
- Equivalent weight = atomic mass ÷ valency. For aluminium: 27 ÷ 3 = 9 g.

7.3 Corrosion and its Prevention

LESSON 36

1. Rusting (Corrosion)

Rusting is iron reacting with oxygen and water to form iron oxide (rust). Machines and iron objects are damaged by rust.

2. Protection from Rusting

- Painting the surface.
- Greasing or oiling.
- Galvanising — coating iron with zinc.
- Electroplating with another metal.

3. Electroplating

Electroplating deposits a thin layer of one metal on another object using electricity. The steps are:

- **Step 1:** The object to be plated is cleaned.
- **Step 2:** It is made the cathode.
- **Step 3:** The plating metal is made the anode.
- **Step 4:** Both are dipped in an electrolyte (a solution of the plating metal's salt) and current is passed, so the metal deposits on the cathode (the object).

4. Examples

- **Tin or zinc on iron:** the iron object is made the cathode and tin or zinc the anode; both are dipped in a solution of that metal's salt and current is passed, so tin or zinc deposits on the iron.
- **Silver plating:** the object is made the cathode and silver the anode; both are dipped in a solution of a silver salt (such as silver nitrate) and current is passed, so silver deposits on the object.

7.4 Alloy Formation

LESSON 37

1. What is an Alloy?

An **alloy** is a homogeneous mixture of two or more metals (a solid-in-solid solution). Alloys are made to improve the properties of metals, such as strength and resistance to rusting.

2. Common Alloys

Alloy	Composition
Brass	Cu–Zn (copper + zinc)
Bronze	Cu–Sn (copper + tin)
Steel	Fe–C (iron + carbon)
Duralumin	Ni–Mg–Al–Cu
Bell metal	A copper–tin alloy

Chemical Reactivity

UNIT 8

8.1 Metals

LESSON 38

1. Properties of Metals

- Metals are lustrous (shiny), malleable and ductile.
- They are good conductors of heat and electricity.
- They are sonorous and have high melting points.
- Metals give electrons to form positive ions (cations), e.g. Na → Na+ + e−. That is why metals conduct electricity and form ionic compounds.
- Most metals are solids, but mercury (Hg) and gallium (Ga) are found in the liquid state.

2. Alkali Metals (Group IA)

Alkali metals are the very reactive metals of Group IA: lithium (Li), sodium (Na), potassium (K), rubidium (Rb), caesium (Cs), francium (Fr). They are not found free in nature because they are very reactive and react quickly with air and water, so they cannot stay in the free state and occur only in compounds.

3. Alkaline-Earth Metals (Group IIA)

Alkaline-earth metals are the metals of Group IIA: beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), radium (Ra). From the list Sr, Sb, Si, Se, B, Ba, Br, Bi — Ba and Sr are the alkaline-earth metals.

4. Soft Metals and Hard Metals

Property	Sodium (soft metal)	Iron (hard metal)
Mohs hardness	0.5	4.5
Melting point	98°C	1535°C
Boiling point	890°C	2450°C
Density	0.971 g/cm³	7.87 g/cm³

Soft metals can be cut with a knife and have low hardness and melting points, such as Na, K, Rb, Cs. Hard metals have high hardness and melting points and strong metallic bonding, such as Ni, Fe, W.

5. Cations and Anions

- **Cation:** a positive ion, e.g. Na+ and Ca^{2+}.
- **Anion:** a negative ion, e.g. Cl− and O^{2-}.

6. Noble Metals

Noble metals such as gold, silver and platinum do not react easily with other substances, because they are chemically stable and are not affected by air, water and acids. Noble metals: gold (Au), silver (Ag), platinum (Pt), iridium (Ir), osmium (Os), ruthenium (Ru), rhodium (Rh), palladium (Pd).

7. Importance of Silver and Gold

- **Silver:** a very good conductor of heat and electricity; used in jewellery, coins, utensils, photography, electronics and medicines. It is a noble metal and does not rust easily. Silver nitrate (AgNO3) is called lunar caustic.
- **Gold:** a shiny, soft, malleable and rust-free metal; used in jewellery, coins, dentistry, electronics and space technology. An astronaut's helmet visor is coated with gold because gold reflects infrared/solar radiation and protects from heat. Gold is used in mirror glass because it is highly reflective and does not tarnish. In dentistry, gold is used in fillings and crowns because it is inert, non-corrosive and biocompatible.
- **Platinum** is used as a catalyst: it speeds up reactions without being consumed, e.g. in catalytic converters and the Ostwald process.

8. Importance of Sodium and Magnesium

- **Sodium** (Group IA alkali metal): used in salt, soda, soap, glass and chemical industries; it also plays an important role in the body.
- **Magnesium** (Group IIA alkaline-earth metal): important in light strong alloys, fireworks, medicines and as part of chlorophyll in plants. Its uses include alloys, fireworks/flares, medicine (antacid), chlorophyll and sacrificial protection.

9. Uses of Some Compounds

- **Washing soda (Na_2CO_3):** cleaning, water softening, making glass and soap.
- **Bleach ($Ca(OCl)_2$):** disinfectant and whitening.
- **Sodium nitrate ($NaNO_3$):** fertiliser, food preservation, glass.
- **Lime:** cement, mortar, whitewash, steel.
  Gypsum/plaster: plaster of Paris, wallboard, cement.
  Calcium tungstate: fluorescent lamps, X-ray screens.

10. Classification Practice

From Ge, Ba, P, K, Si, Sr, S, Sb, Ca, C — metalloids: Ge, Si, Sb; metals: Ba, K, Sr, Ca; non-metals: P, S, C.

8.2 Non-Metals

LESSON 39

1. Properties of Non-Metals

- Non-metals are dull (not shiny) and brittle.
- They are poor conductors of electricity.
- They have low melting and boiling points.
- They gain electrons to form anions, and form acidic oxides.
- Non-metals attract electrons strongly, so their electronegativity is high. Fluorine is the most electronegative element. They lack metallic properties: not shiny, not conductors, brittle.
- Chlorine (Cl) acts as an oxidising agent in a chemical reaction because it takes electrons and oxidises another substance.

2. Halogens (Group VIIA)

Halogens are the reactive non-metals of Group VIIA: fluorine (F), chlorine (Cl), bromine (Br), iodine (I), astatine (At). They are very reactive because their outer shell has 7 electrons and taking one electron completes the octet. Reactivity decreases from top to bottom: F > Cl > Br > I.

- Only fluorine, being above chlorine and more reactive, can displace chlorine.
- Iodine (I_2) is the halogen found in the solid state; bromine (Br_2) is a liquid; F_2 and Cl_2 are gases.
- Acidity of halogen acids increases in the order: HF < HCl < HBr < HI, i.e. HF is the weakest and HI is the strongest acid.

3. Noble Gases (Group VIIIA)

The non-metals of Group VIIIA are found in the gaseous state — the noble gases (He, Ne, Ar, Kr, Xe, Rn). From N, Na, Ni, Ne, Ar, At, He — the Group VIIIA elements are Ne, Ar and He. Group VIIIA is the group that has only non-metals in the gaseous state.

4. Metalloids

Metalloids show properties between metals and non-metals. The book's metalloid list is: B, Si, Ge, As, Sb, Te, Po, At. (Note: in the book's question, Se is the closest choice classed as a metalloid in some references.) Any three: boron (B), silicon (Si), germanium (Ge).

5. Group VA Non-Metals

Nitrogen (N), phosphorus (P), arsenic (As), antimony (Sb), bismuth (Bi).

6. Tests and Special Facts

- **CO_2 test:** carbon dioxide turns limewater milky because it combines with limewater to form calcium carbonate.
- **Chloride test (Cl^-):** adding silver nitrate to the solution gives a white precipitate.
- **Iodide test (I^-):** adding silver nitrate gives a yellow precipitate.